

ADL Homework 4 — Convolutional Neural Networks on CIFAR-10

Applied Deep Learning, Spring 2026

We implement the PyTorch CIFAR-10 tutorial CNN, train it for image classification, and (in later tasks) extend it with a deconvolutional decoder for reconstruction and latent-feature analysis.

Hyperparameters (Task 1)

We follow the [PyTorch CIFAR-10 tutorial](#) and the Lecture 7 demo. Because computational resources are limited, we train on **6,000 images** (one tenth of the CIFAR-10 training set), as allowed by the assignment.

Parameter	Value
Architecture	Conv(3→6, k=5) → ReLU → MaxPool → Conv(6→16, k=5) → ReLU → MaxPool → FC(400→120) → ReLU → FC(120→84) → ReLU → FC(84→10)
Loss	Cross-entropy
Optimizer	SGD (lr = 0.001, momentum = 0.9)
Batch size	64
Epochs	40
Training subset	6,000 images (random sample, seed 42)
Test set	10,000 images (full CIFAR-10 test set)
Input normalization	ToTensor + normalize to [-1, 1] with mean/std (0.5, 0.5, 0.5)

Device: cpu
Training on 6000 images for 40 epochs
Task 2 lambda: 0.1

Task 1 — CIFAR-10 classification

We train the CNN below and report training/test accuracy, learning curves, sample test predictions, and per-class accuracy.

Epoch 01/40		train loss 2.305	acc 10.2%		test loss 2.304	acc 10.3%
Epoch 10/40		train loss 2.224	acc 17.8%		test loss 2.202	acc 17.9%
Epoch 20/40		train loss 1.853	acc 32.7%		test loss 1.866	acc 33.0%
Epoch 30/40		train loss 1.591	acc 41.7%		test loss 1.598	acc 41.8%
Epoch 40/40		train loss 1.461	acc 46.6%		test loss 1.535	acc 44.4%

Final train accuracy: 46.6%

Final test accuracy: 44.4%

Epoch		train loss		train acc (%)		test loss		test acc (%)
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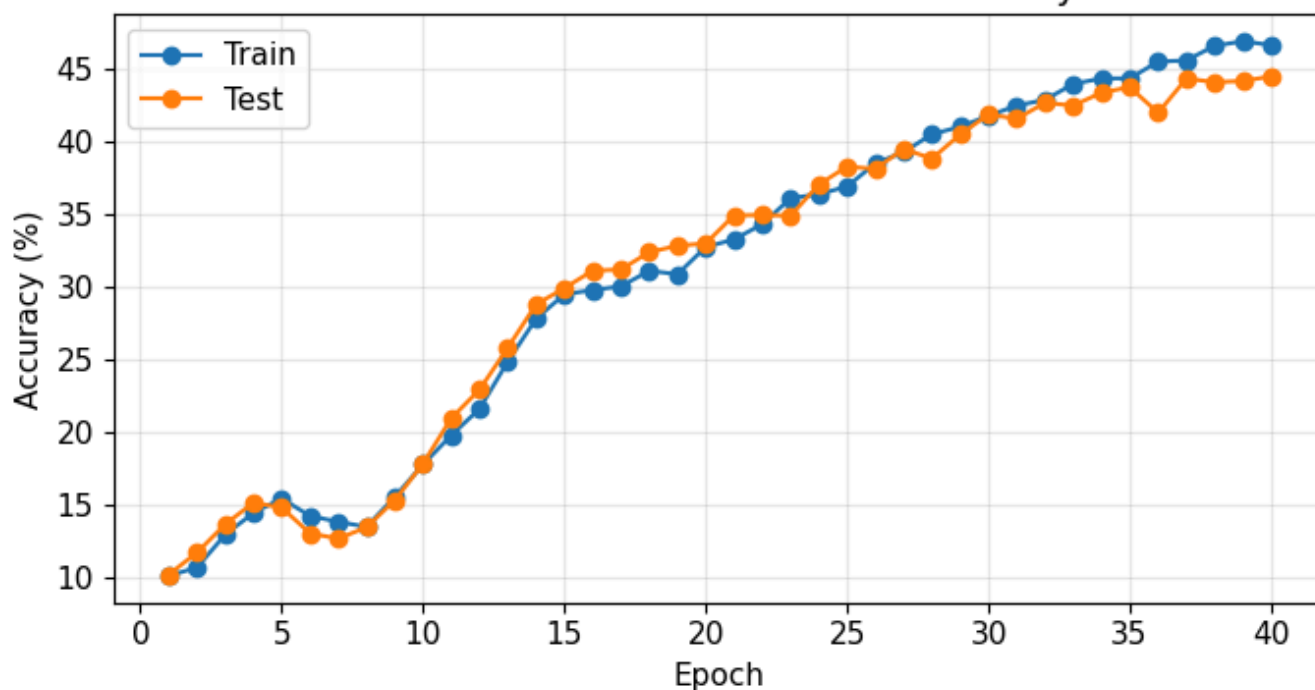
1		2.305		10.2		2.304		10.3
10		2.224		17.8		2.202		17.9
20		1.853		32.7		1.866		33.0
30		1.591		41.7		1.598		41.8
40		1.461		46.6		1.535		44.4

Per-class test accuracy:

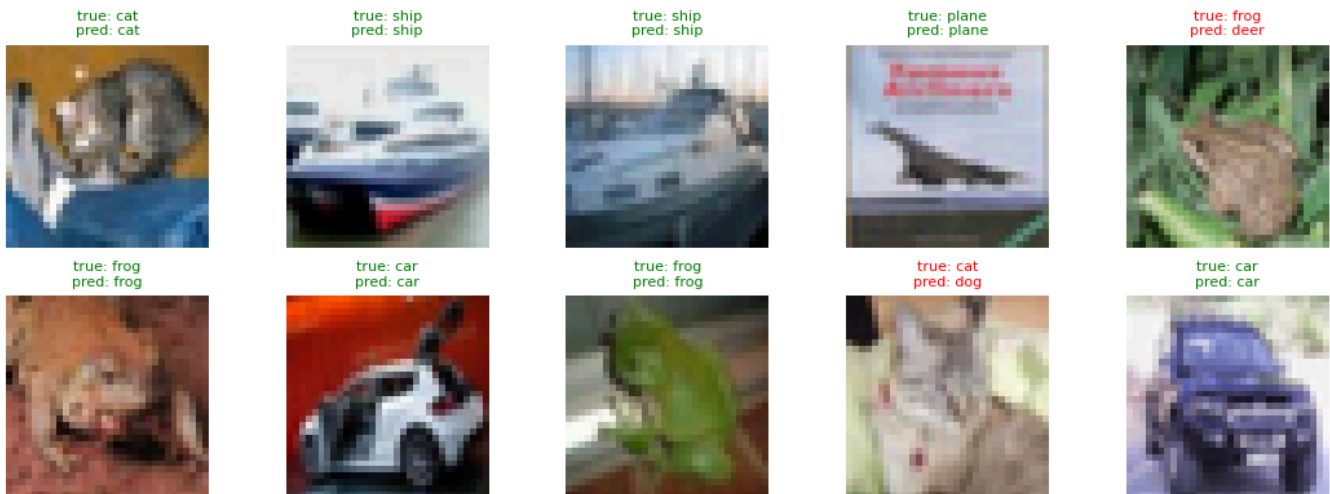
Class		Accuracy (%)
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plane		53.5
car		52.3
bird		35.5
cat		20.1
deer		36.6
dog		59.0
frog		40.2
horse		46.8
ship		46.3
truck		54.1

Task 1: CIFAR-10 classification accuracy



Test images with predicted labels



Discussion (Task 1)

Training on **6,000 images** for **40 epochs** reached **46.6% train** and **44.4% test** accuracy. Learning was slow at first (17.9% test at epoch 10), then improved steadily to 44.4% by epoch 40. Train and test track closely, so there is little overfitting — the main limitation is the small training subset.

Per-class accuracy is uneven: **dog** (59.0%), **truck** (54.1%), and **plane** (53.5%) are strongest, while **cat** (20.1%) and **bird** (35.5%) are weakest. The CNN captures coarse shape cues for vehicles more easily than fine-grained animal classes.

Task 2 — Deconvolutional model

We extend the Task 1 CNN with a decoder that reconstructs the input image using **max-unpooling** and **transposed convolutions**. The network is trained with

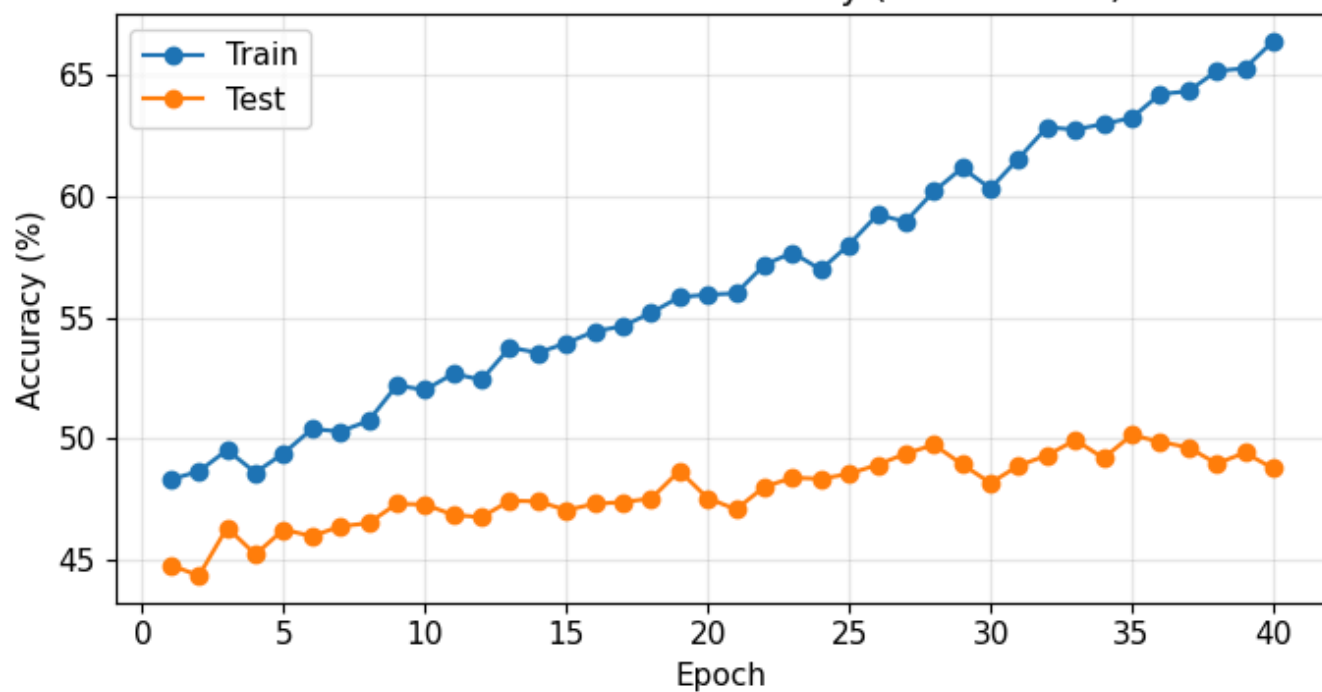
$$\mathcal{L} = \mathcal{L}_{ce} + \lambda \mathcal{L}_{rec},$$

where $\mathcal{L}_{rec} = \frac{1}{3} \sum_{i=1}^3 \|\tilde{x}_i - x_i\|_F^2$ is the mean per-channel MSE over RGB. We set $\lambda = 0.1$: large enough to produce visible reconstructions, but small enough that classification accuracy remains close to the classifier-only model. The encoder and classifier are initialized from the trained Task 1 weights; the decoder is trained from scratch.

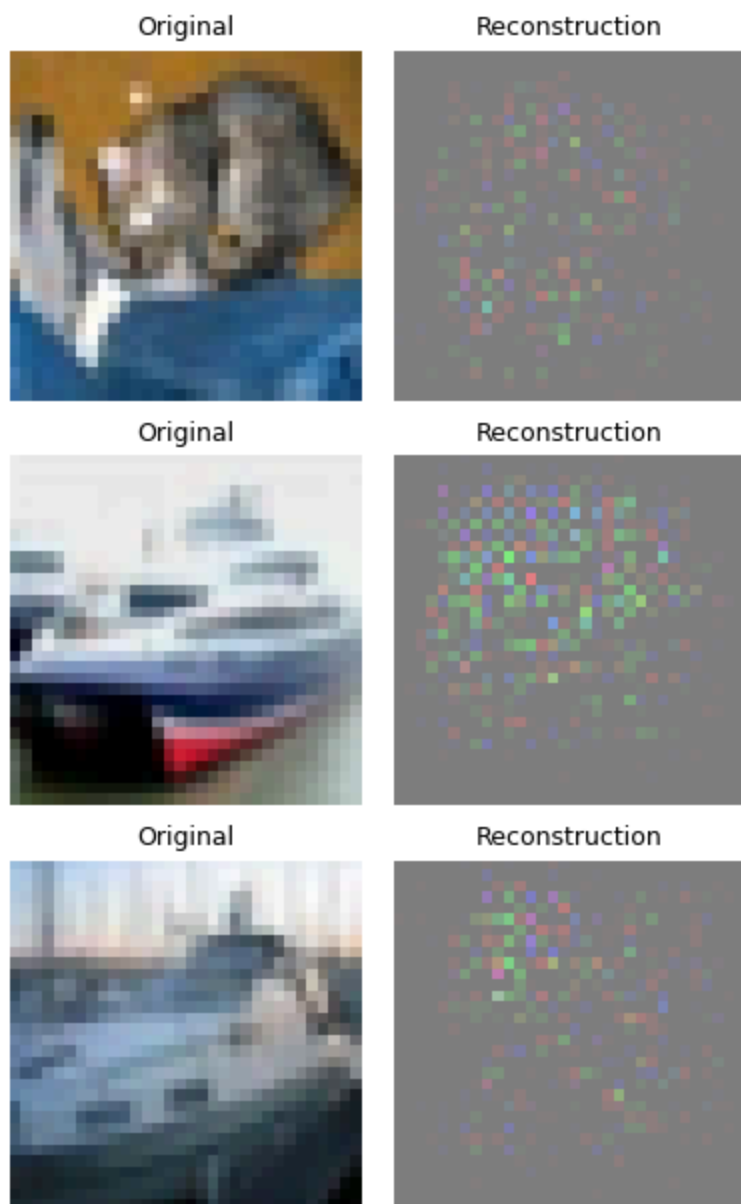
Epoch 01/40	train acc 48.3%	test acc 44.8%	train rec 0.2833	test rec 0.2780
Epoch 10/40	train acc 52.0%	test acc 47.3%	train rec 0.2622	test rec 0.2606
Epoch 20/40	train acc 56.0%	test acc 47.5%	train rec 0.2581	test rec 0.2566
Epoch 30/40	train acc 60.3%	test acc 48.2%	train rec 0.2565	test rec 0.2550
Epoch 40/40	train acc 66.4%	test acc 48.8%	train rec 0.2553	test rec 0.2539

Final test accuracy (Task 2): 48.8%
Lambda: 0.1

Task 2: classification accuracy (lambda=0.1)



Task 2: original vs reconstructed test images



Discussion (Task 2)

With $\lambda = 0.1$ and weights initialized from Task 1, joint training reached **66.4% train** and **48.8% test** accuracy (peak **50.1%** at epoch 35), improving test accuracy by ~ 4 points over Task 1. Test reconstruction MSE decreased from **0.278 to 0.254**, but the reconstructions remain blurry, as expected for a shallow decoder on 32×32 images.

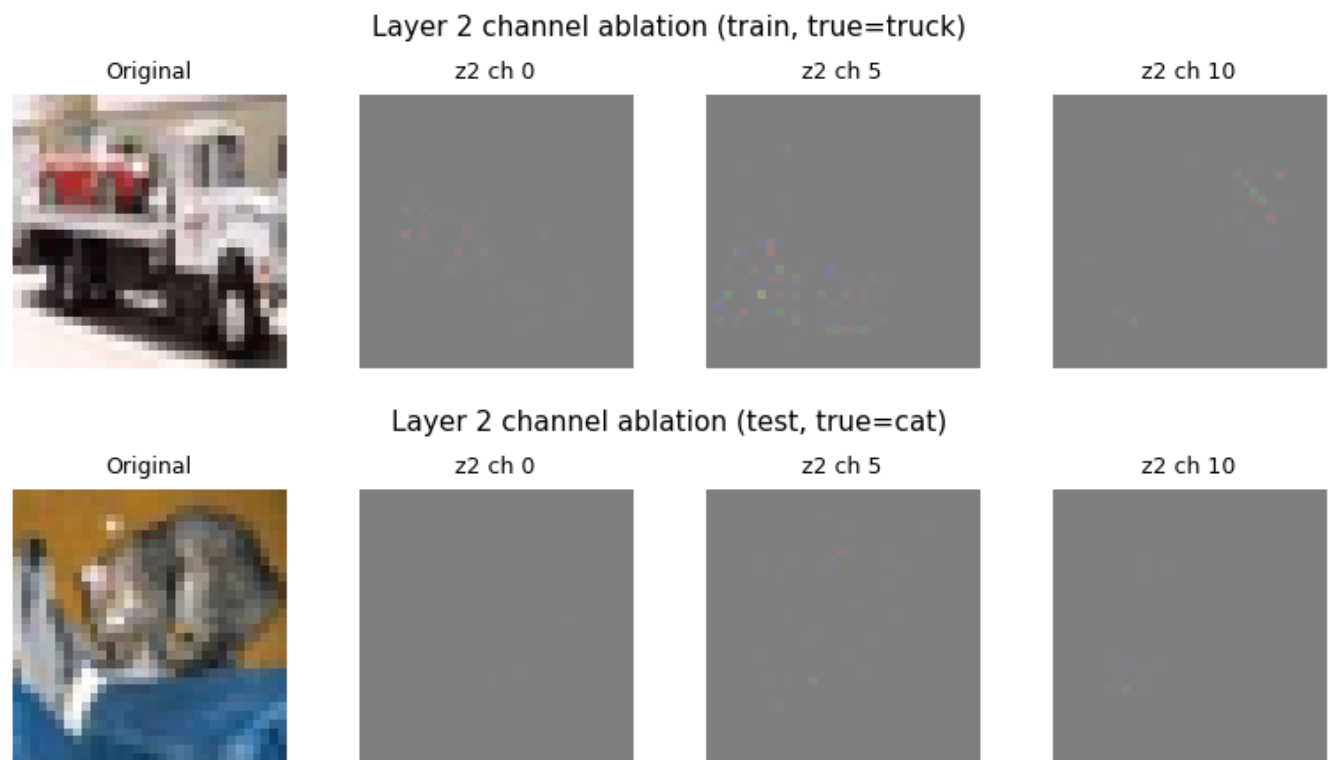
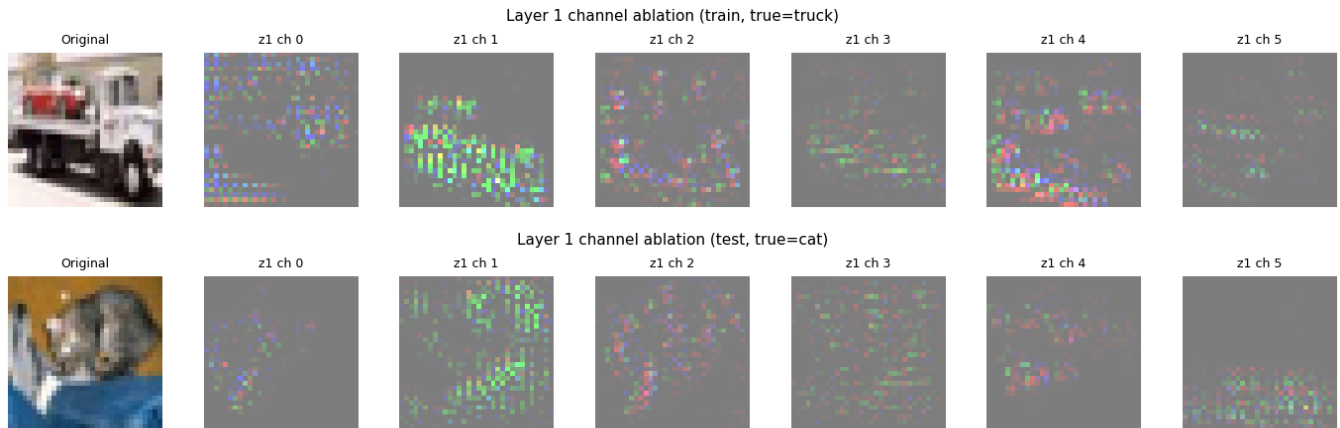
A train-test gap opens (66.4% vs 48.8%), suggesting mild overfitting once the decoder is added. We used $\lambda = 0.1$ to balance visible reconstructions with classification performance.

Task 3 — Latent representation analysis

We select one **training** image and one **test** image, then reconstruct from isolated channels of $z^{(1)}$ (all 6 channels) and $z^{(2)}$ (channels 0, 5, and 10). For each channel, all other channels are set to zero before decoding.

Train image index 0: true class = truck

Test image index 0: true class = cat



Discussion (Task 3)

We used train index 0 (**truck**) and test index 0 (**cat**).

Layer 1 ($z^{(1)}$). Individual channels produce localized edge/color/texture patterns. On the truck, channel 1 highlights the lower-left (wheels/chassis) and channel 4 the red car; on the cat, channel 1 covers body texture and channel 5 the bottom edge.

Layer 2 ($z^{(2)}$, channels 0, 5, 10). Reconstructions are much sparser. Channel 0 is blank for both images; channels 5 and 10 show only small pixel clusters (wheel/rear on truck, faint specks on cat). Deeper features are distributed — no single channel reconstructs a recognizable part alone.

Summary. $z^{(1)}$ channels act as local feature detectors; $z^{(2)}$ requires many channels together for a coherent reconstruction.